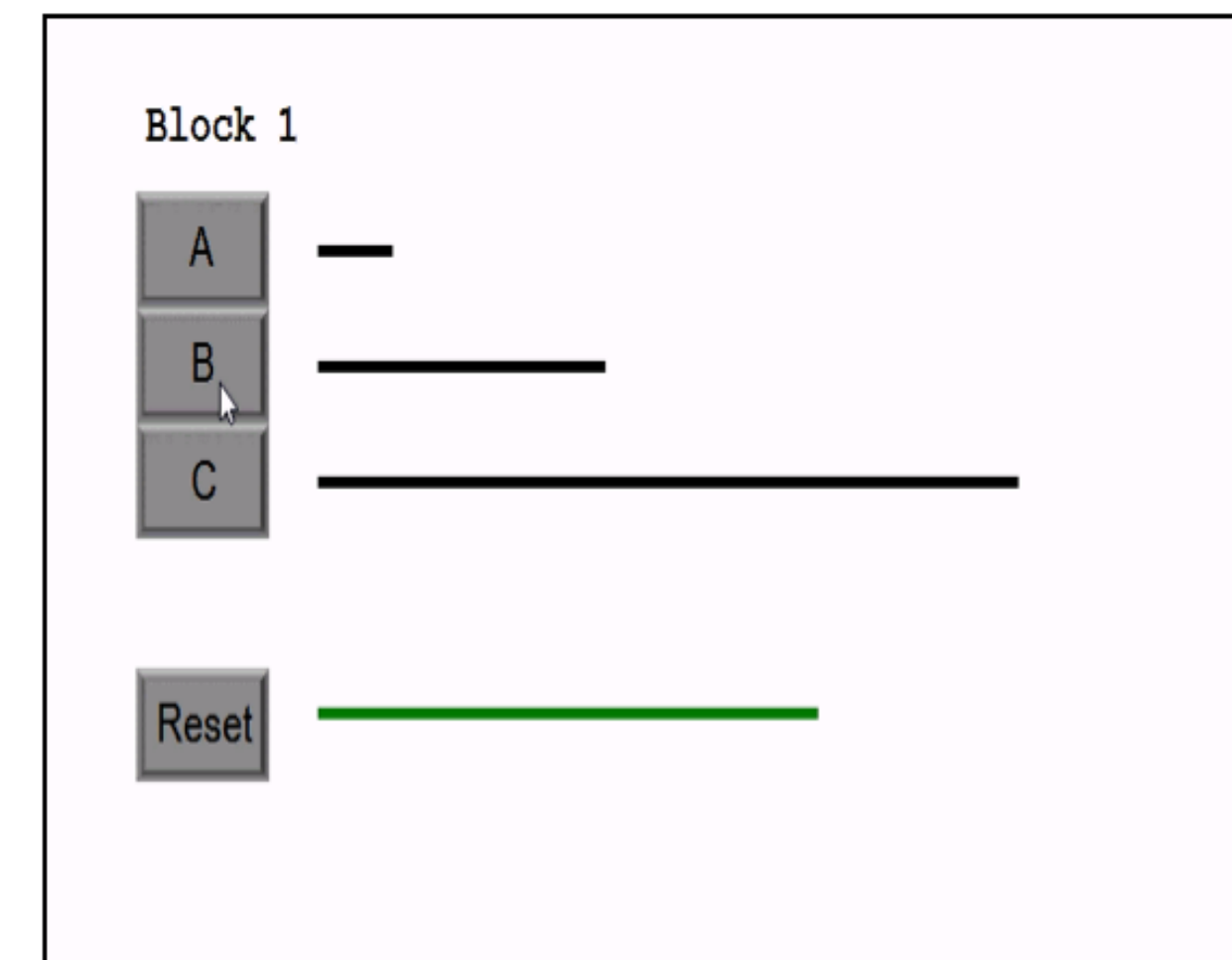


Introduction

- Individual differences in working memory span are related to successfully solving problems (Ash & Wiley, 2006).
- Problem space refers to the mental representation of a problem and of all the possible paths to solving it.
- The Building Sticks Task (BST) is a simple problem-solving task in which problems can be solved quickly within five moves but with a problem space containing 40-60 distinct states.
 - Three building sticks and a target stick are presented as shown in the figure.
 - The goal is to use the three sticks to create another stick that matches the length of the target stick.
 - Participants were forced to restart the problem after using more than 6 moves. Otherwise, they could use a "reset" button to start a new solution attempt.
- Problem-solving in this task consists of a sequence of attempts to solve the problem until the solution is found. Across these attempts, participants sometimes repeat an attempt from earlier, so there can be unique and repeated attempts while solving a problem.
- We used this task to investigate the relationship between working memory capacity (WMC) and problem space exploration.
- WMC was measured by two complex span tasks: Orientation Span (Shah & Miyake, 1996) and Reading Span (Turner & Engle, 1989).
- Proportion of unique attempts and forced resets are measures of problem space exploration.

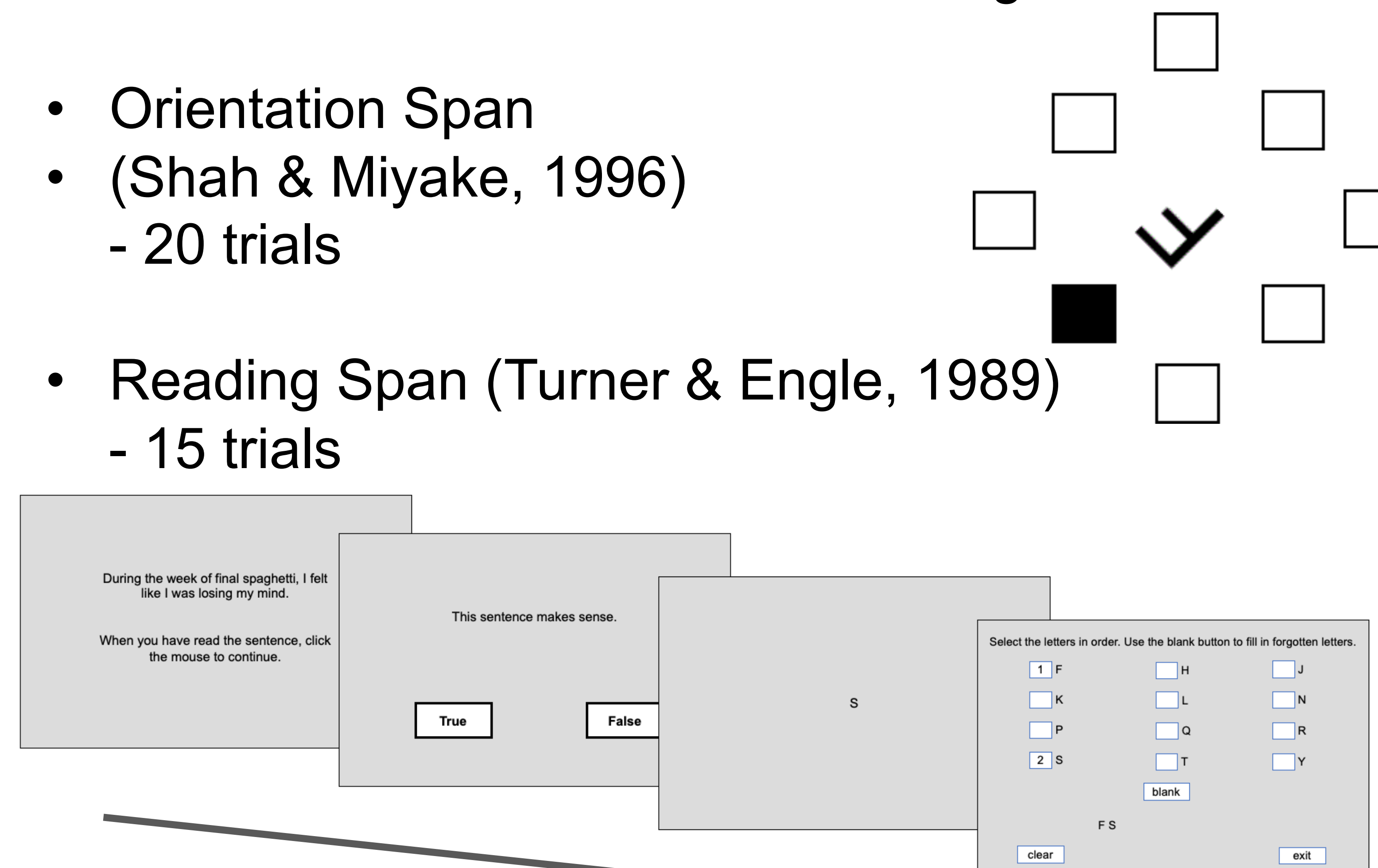


Methods

Participants: 147 undergraduates from MSU, only 101 left after excluding the outliers of working memory measures

Tasks:

- BST (Schunn et al., 2001)
 - 7 blocks with each block consisting of 10 trials
- Orientation Span (Shah & Miyake, 1996)
 - 20 trials
- Reading Span (Turner & Engle, 1989)
 - 15 trials

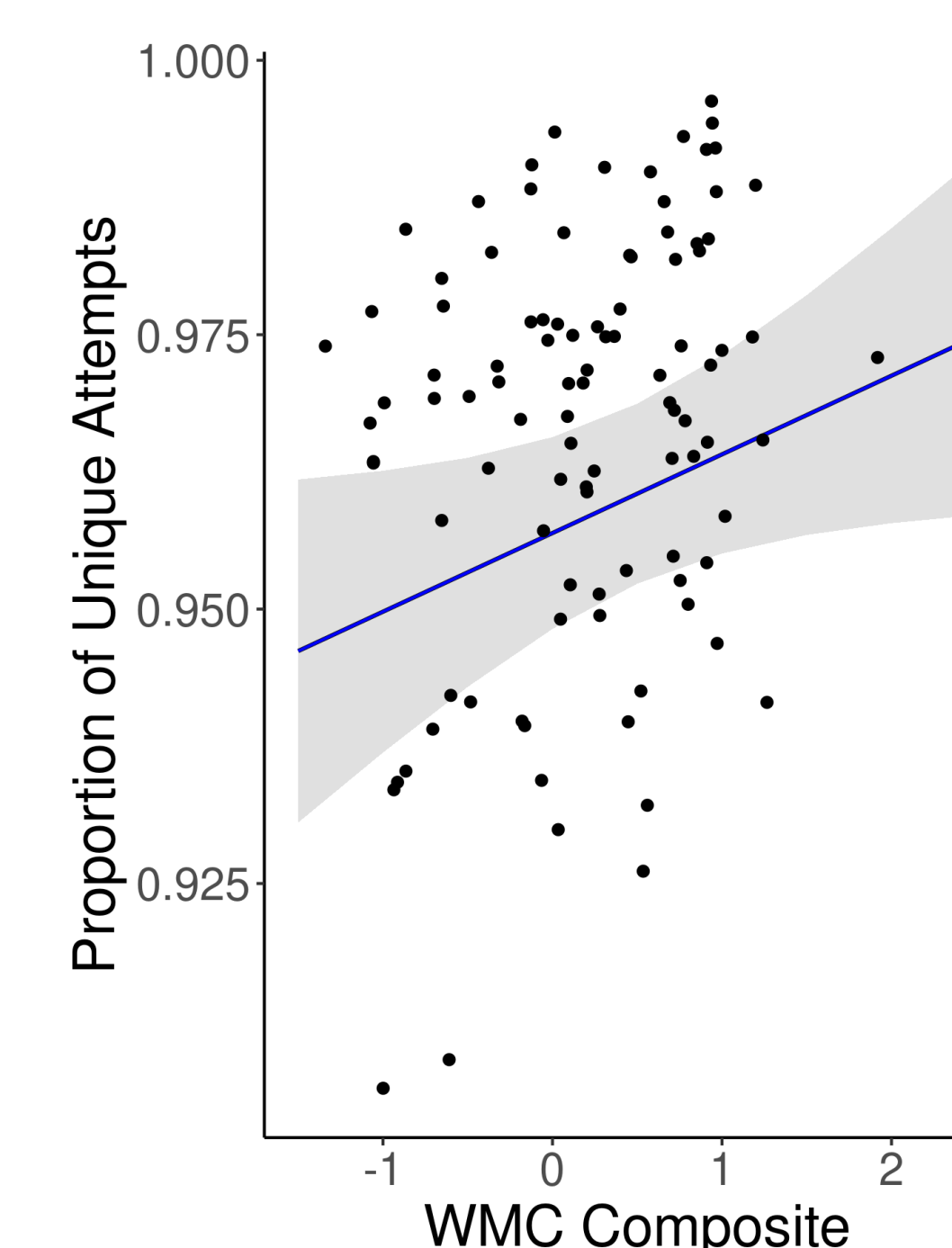


Procedure:
BST(about 10 min) – RSPAN (about 10 min) – OSPAN (about 15 min)

Results

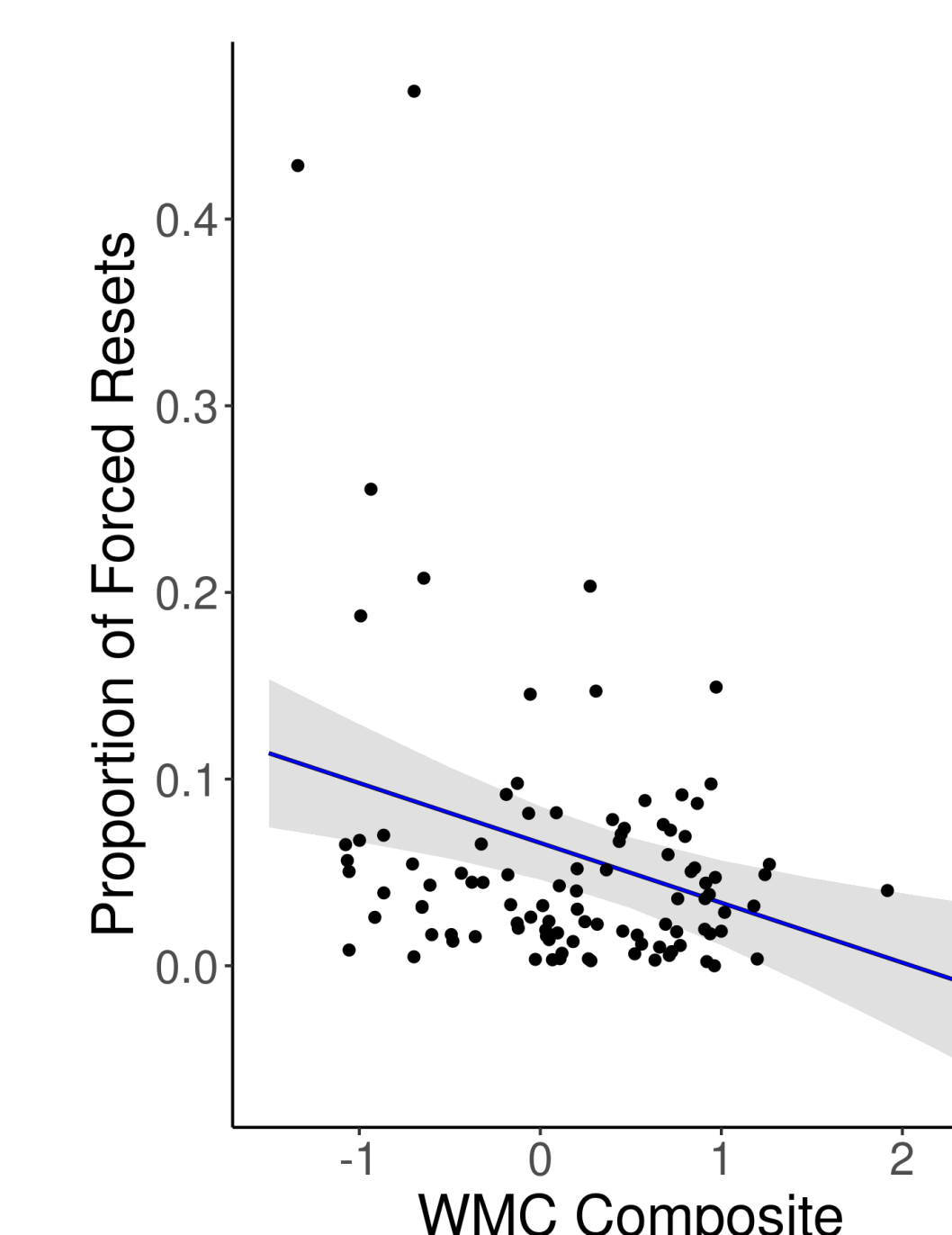
1. WMC and unique attempts

Individuals with higher WMC have more unique attempts, $t(97.39) = 2.05$, $p < .05$.



2. WMC and forced resets

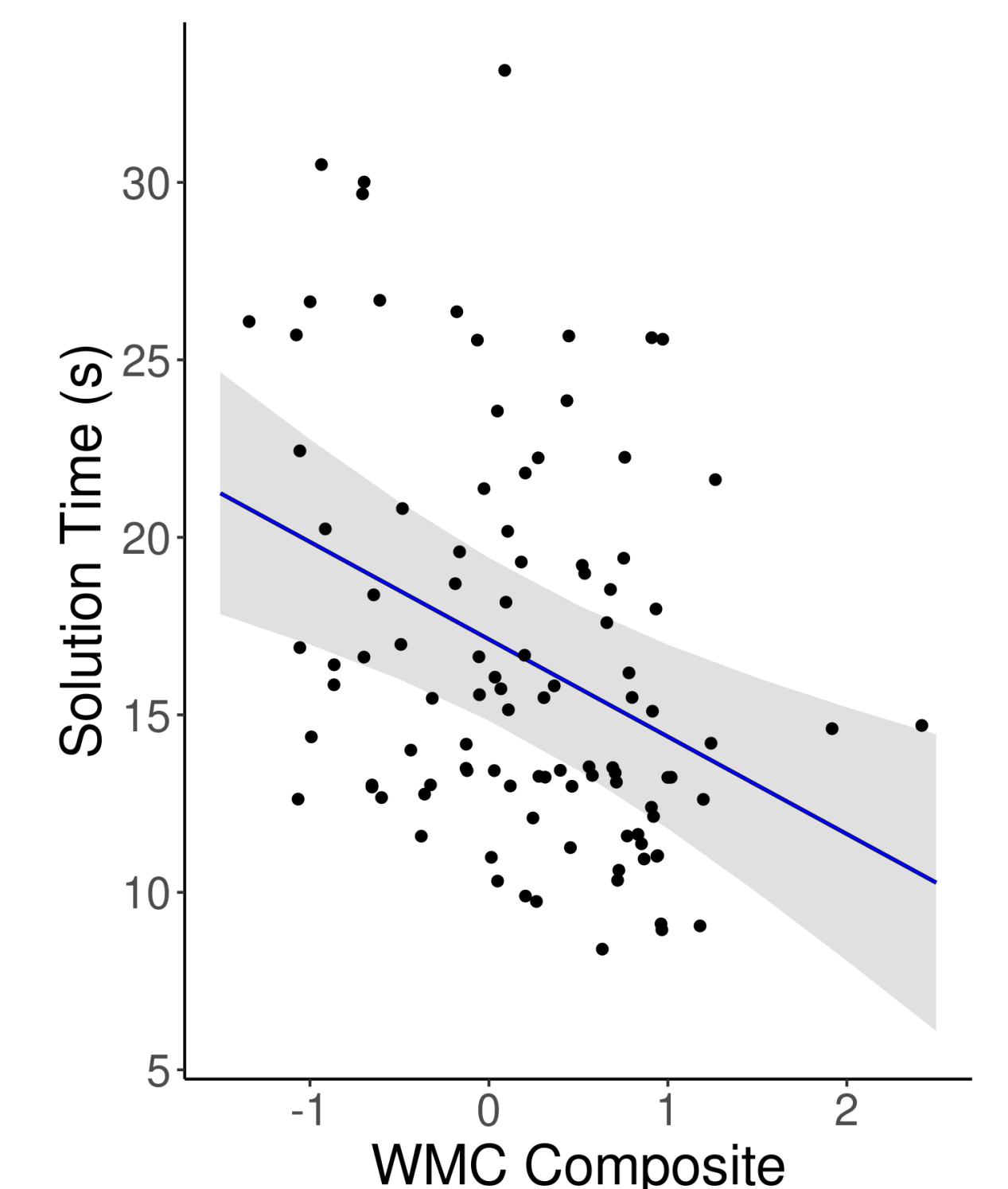
Individual with higher WMC have fewer forced resets, $t(104.47) = -3.27$, $p < .01$.



Results

3. WMC and solution time

Individuals with higher WMC solve the problems faster, $t(98.77) = -3.56$, $p < .001$.



4. There is no significant effect of WMC on solution time after including unique attempts and forced resets in the model.

Discussion

- WMC predicts more effective problem space exploration.
 - Higher WMC was associated with faster solution time in attempt to solving problems.
 - Higher WMC was associated with a higher proportion of unique attempts in problem solving.
 - Higher WMC was also associated with more self-terminated solution attempts as opposed to being forced to restart the problem after six moves by the system.
- There is a mediation of the relationship between WMC and solution time. One interpretation is that WMC affects these measures of problem space exploration which in turn influence solution time.

Acknowledgement & References

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➤ Research Questions:

How are individual differences in WMC associated with problem space exploration?